

Fusion systems in representation theory

Three lectures at the University of Valencia

Benjamin Sambale*

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1 Fusion in groups

Definition 1.1. Let $H \leq G$ be finite groups. Elements $x, y \in H$ (or subsets) are called *fused* in G if they are conjugate in G , but not in H .

Example 1.2.

- (i) The permutations $(123), (132) \in A_3$ are fused in S_3 .
- (ii) Let $X, Y \leq H$ be isomorphic subgroups via an isomorphism $\varphi: X \rightarrow Y$. We embed H into $G := \text{Sym}(H)$ via the regular representation $\sigma: H \rightarrow G, h \mapsto \sigma_h$ where $\sigma_h(g) = hg$ for $g, h \in H$. Let $\{r_1, \dots, r_n\}$ and $\{s_1, \dots, s_n\}$ be systems of representatives for the right cosets in H of X and Y , respectively.¹ Let $\hat{\varphi} \in G$ with $\hat{\varphi}(xr_i) := \varphi(x)s_i$ for all $x \in X$ and $1 \leq i \leq n$. For $x \in X, y \in Y$ and $1 \leq i \leq n$ we have

$$(\hat{\varphi}\sigma_x\hat{\varphi}^{-1})(ys_i) = (\hat{\varphi}\sigma_x)(\varphi^{-1}(y)r_i) = \hat{\varphi}(x\varphi^{-1}(y)r_i) = \varphi(x)ys_i = \sigma_{\varphi(x)}(ys_i)$$

and $\hat{\varphi}\sigma_x\hat{\varphi}^{-1} = \sigma_{\varphi(x)}$. Hence, φ is realized by the conjugation with $\hat{\varphi}$ in G .²

- (iii) A consequence of (ii) is that elements $x, y \in H$ of the same order are conjugate in some finite group $G \geq H$.

Goal: Find “small” subgroups $K \supseteq H$ *controlling fusion* in H , i. e. $x, y \in H$ are fused in G if and only if x, y are fused in K .

Main interest: $H \in \text{Syl}_p(G)$.

In the following let $P \in \text{Syl}_p(G)$. Let $\text{O}_{p'}(G)$ be the largest normal p' -subgroup of G . If no elements of P are fused in G , then G is called *p-nilpotent*.

Theorem 1.3 (FROBENIUS). *The following assertions are equivalent:*

- (1) G is *p-nilpotent*.

*Leibniz Universität Hannover, Germany, sambale@math.uni-hannover.de

¹Using Hall’s marriage theorem one can show that a *common* system of representatives exists for these right cosets. This is not needed here.

²This construction fails for infinite groups since for example the isomorphism $\mathbb{Z} \rightarrow 2\mathbb{Z}$ does not extend to $\mathbb{Z} \rightarrow \mathbb{Z}$. In those situations one can use HNN-extensions.

(2) $N_G(Q)/C_G(Q)$ is a p -group for all $Q \leq P$.

(3) $G = O_{p'}(G)P$.

Example 1.4. Every p' -group and every nilpotent group is p -nilpotent.

Theorem 1.5 (BURNSIDE). $N_G(P)$ controls fusion in $Z(P)$.

Proof. Let $x, y \in Z(P)$ and $g \in G$ such that ${}^g x := gxg^{-1} = y$. Then $P \leq C_G(y)$ and ${}^g P \leq {}^g C_G(x) = C_G({}^g x) = C_G(y)$. By Sylow's theorem, there exists $c \in C_G(y)$ such that ${}^{cg} P = P$. Now $h := cg \in N_G(P)$ such that ${}^h x = {}^c({}^g x) = {}^c y = y$. $\square$

Theorem 1.6 (Z^* -theorem³). If $z \in Z(P)$ is not fused to any other element in P , then $G = O_{p'}(G)C_G(z)$.

Proof. Glauberman proved the theorem for $p = 2$ using representation theory, while the only known proof for $p > 2$ is via the classification of finite simple groups (CFSG for short). $\square$

Example 1.7. If P is a (generalized) quaternion 2-group, then $G = O_{2'}(G)C_G(Z(P))$ since $Z(P)$ is generated by the unique involution in P .⁴

Goldschmidt and Flores–Foote classified more generally groups G with $A \trianglelefteq P$ such that no element of A is fused to an element of $P \setminus A$ (i.e. A is *strongly closed* in P). Let

$$J(P) := \langle A \leq P : A \text{ abelian of maximal order} \rangle$$

be the *Thompson subgroup* of P .⁵

Theorem 1.8 (THOMPSON). If $p \geq 5$, then G is p -nilpotent if and only if $N_G(J(P))/C_G(J(P))$ is a p -group.

Theorem 1.9 (GLAUBERMAN'S ZJ-theorem). Let $p > 2$. Then G is p -nilpotent if and only if $N_G(Z(J(P)))$ is p -nilpotent. If G has no section isomorphic to $Qd(p) := C_p^2 \rtimes \text{SL}_2(p)$, then $N_G(Z(J(P)))$ controls fusion in P .

Example 1.10. For $p \geq 5$, every (p) -solvable group is $Qd(p)$ -free.

Theorem 1.11 (STELLMACHER). If $p = 2$ and G has no section isomorphic to $Qd(2) \cong S_4$, then $N_G(W)$ controls fusion in P for some characteristic subgroup W of P . If $P \neq 1$, then $W \neq 1$.

Let $G' = [G, G]$ be the commutator subgroup and $O^p(G) = \langle p'\text{-elements} \rangle$ the p -residue of G .

Theorem 1.12 ((Hyper)focal subgroup theorem).

$$\text{foc}_G(P) := \langle xy^{-1} : x, y \in P \text{ are conjugate in } G \rangle = G' \cap P \quad (\text{focal subgroup}),$$

$$\text{hyp}_G(P) := \langle xy^{-1} : x, y \in P \text{ are conjugate by a } p'\text{-element} \rangle = O^p(G) \cap P \quad (\text{hyperfocal subgroup}).$$

³It is often assumed that z has order p , but this is unnecessary.

⁴This special case of the Z^* -theorem was first proved by Brauer–Suzuki.

⁵Several non-equivalent definitions of the Thompson subgroup are used in the literature.

The *transfer map* yields $G/O^p(G) \cong P/\mathfrak{hnp}_G(P)$.

Theorem 1.13 (GRÜN's theorem).

$$\mathfrak{foc}_G(P) = [N_G(P), P] \langle P \cap Q' : Q \in \text{Syl}_p(G) \rangle.$$

Let $\Phi(P)$ be the Frattini subgroup of P .

Theorem 1.14. *The following assertions are equivalent:*

- (1) G is p -nilpotent.
- (2) $\mathfrak{hnp}_G(P) = 1$.
- (3) $\mathfrak{hnp}_G(P) \leq \Phi(P)$.

Theorem 1.15 (TATE's transfer theorem). *For $P \leq H \leq G$ we have*

$$\mathfrak{foc}_G(P) = \mathfrak{foc}_H(P) \iff \mathfrak{hnp}_G(P) = \mathfrak{hnp}_H(P) \iff \mathfrak{foc}_G(P)\Phi(P) = \mathfrak{foc}_H(P)\Phi(P).$$

If $\mathfrak{foc}_G(P) = \mathfrak{foc}_H(P)$, we say that H *controls transfer* in P . In this case H determines whether G is p -nilpotent by Theorem 1.14.

Theorem 1.16 (YOSHIDA's transfer theorem). *If P has no quotient isomorphic to $C_p \wr C_p$, then $N_G(P)$ controls transfer in P .*

Example 1.17.

- (i) If $|P| \leq p^p$ or $\exp(P) = p$ (exponent) or $c(P) < p$ (nilpotency class), then $N_G(P)$ controls transfer in P . This follows from the properties of $C_p \wr C_p$.
- (ii) Let $p = 2$ and $G = S_4$. Then $N_G(P) = P \cong D_8 \cong C_2 \wr C_2$ does not control transfer in P since otherwise G would be 2-nilpotent. For $p > 2$ and

$$G = \mathbb{F}_p^p \rtimes \left\langle \begin{pmatrix} 1 & & & 0 \\ & -1 & & \\ & & \ddots & \\ 0 & & & -1 \end{pmatrix}, \begin{pmatrix} 0 & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 1 & & & 0 \end{pmatrix} \right\rangle \leq \text{ASL}(p, p),$$

again $N_G(P) = P \cong C_p \wr C_p$ does not control transfer in P .

Theorem 1.18 (GLAUBERMAN). *If $p \geq 5$, then there exists a characteristic subgroup K of P such that $N_G(K)$ controls transfer in P and $Z(P) \leq K$.*

The simple group $\text{PSL}(2, 17)$ shows that Theorem 1.18 fails for $p = 2$ (here P is a maximal subgroup). It is an open problem whether Theorem 1.18 holds for $p = 3$. For $p \geq 7$ one can take $K = J(P)$.

2 Fusion systems

For arbitrary groups $S, T \leq P$ let $\text{Hom}_P(S, T)$ be the set of homomorphisms $S \rightarrow T$ induced by inner automorphisms of P , i. e.

$$\text{Hom}_P(S, T) := \{\varphi: S \rightarrow T : \exists g \in P : \varphi(s) = {}^g s \ \forall s \in S\}.$$

Definition 2.1 (PUIG⁶). A *fusion system* on a finite p -group P is a category $\mathcal{F}$ with objects $\text{Obj}(\mathcal{F}) = \{S : S \leq P\}$ and morphisms $\text{Hom}_{\mathcal{F}}(S, T) \subseteq \{S \rightarrow T : \text{injective group homomorphism}\}$ such that

- $\text{Hom}_P(S, T) \subseteq \text{Hom}_{\mathcal{F}}(S, T)$ for $S, T \leq P$,
- $\varphi \in \text{Hom}_{\mathcal{F}}(S, T) \implies \varphi \in \text{Hom}_{\mathcal{F}}(S, \varphi(S)), \varphi^{-1} \in \text{Hom}_{\mathcal{F}}(\varphi(S), S)$.

Example 2.2.

- Let P be a p -subgroup of a finite group G . Then $\text{Hom}_{\mathcal{F}}(S, T) := \text{Hom}_G(S, T)$ for $S, T \leq P$ defines a fusion system on P , which we denote by $\mathcal{F}_P(G)$. In particular, there is always the *trivial* fusion system $\mathcal{F}_P(P)$, which is a subcategory of every fusion system on P .
- The *universal* fusion system $\mathcal{F} := \mathcal{U}(P)$ on P is defined by

$$\text{Hom}_{\mathcal{F}}(S, T) := \{S \rightarrow T \text{ injective homomorphism}\}.$$

Every fusion system on P is a subcategory of $\mathcal{U}(P)$.

Theorem 2.3 (PARK). *For every fusion system $\mathcal{F}$ on P there exists a finite group G containing P such that $\mathcal{F} = \mathcal{F}_P(G)$.*

Theorem 2.3 remains true even for arbitrary finite groups P with appropriate definitions (see Example 1.2(ii) for $\mathcal{F} = \mathcal{U}(P)$).

Definition 2.4. Let $\mathcal{F}$ be a fusion system on P and $S, T \leq P$.

- S, T are called *$\mathcal{F}$ -conjugate* if there exists an isomorphism $\varphi: S \rightarrow T$ in $\mathcal{F}$.
- S is called *$\mathcal{F}$ -automized* if $\text{Aut}_P(S) \in \text{Syl}_p(\text{Aut}_{\mathcal{F}}(S))$.
- S is called *$\mathcal{F}$ -centralized*⁷ if $|\text{C}_P(S)| \geq |\text{C}_P(T)|$ for all $\mathcal{F}$ -conjugates T of S .
- S is called *$\mathcal{F}$ -normalized* if $|\text{N}_P(S)| \geq |\text{N}_P(T)|$ for all $\mathcal{F}$ -conjugates T of S .
- For an isomorphism $\varphi: S \rightarrow T$ let N_{φ} be the preimage of $\text{Aut}_P(S) \cap \varphi^{-1} \text{Aut}_P(T) \varphi$ under the conjugation map $\text{N}_P(S) \rightarrow \text{Aut}_P(S), x \mapsto c_x$, i. e.

$$N_{\varphi} := \{x \in \text{N}_P(S) : \varphi c_x \varphi^{-1} \in \text{Aut}_P(T)\}.$$

- T is called *$\mathcal{F}$ -receptive* if every isomorphism $\varphi: S \rightarrow T$ in $\mathcal{F}$ extends to N_{φ} (note that $SC_P(S) \leq N_{\varphi} \leq \text{N}_P(S)$).

⁶Puig calls them *Frobenius categories*

⁷often called *fully $\mathcal{F}$ -centralized/normalized*

Example 2.5.

- (i) If $S, T \leq P \leq G$ are fused in G , then they are $\mathcal{F}_P(G)$ -conjugate.
- (ii) If $P \in \text{Syl}_p(G)$, then P is automized in $\mathcal{F}_P(G)$, because $PC_G(P)/C_G(P) \in \text{Syl}_p(N_G(P)/C_G(P))$.
- (iii) Every central subgroup of P is $\mathcal{F}$ -centralized and every normal subgroup is $\mathcal{F}$ -normalized.
- (iv) Every $\mathcal{F}$ -receptive subgroup is $\mathcal{F}$ -centralized: Let $T \leq P$ be receptive and $\varphi: S \rightarrow T$ an isomorphism in $\mathcal{F}$. Then φ extends to $\hat{\varphi}: N_\varphi \rightarrow P$. For $s \in S$ and $g \in C_P(S)$ we have $\hat{\varphi}(g)\varphi(s)\hat{\varphi}(g)^{-1} = \hat{\varphi}(gsg^{-1}) = \varphi(s)$ and $\hat{\varphi}(C_P(S)) \leq C_P(T)$. Since morphisms are injective, it follows that $|C_P(S)| \leq |C_P(T)|$.
- (v) Every $\mathcal{F}$ -centralized, $\mathcal{F}$ -automized subgroup $S \leq P$ is $\mathcal{F}$ -normalized. This follows from $|N_P(S)| = |\text{Aut}_P(S)||C_P(S)|$.
- (vi) Let $S := \langle (12)(34) \rangle \leq P := \langle (1234), (13) \rangle \leq G := S_4$ and $\mathcal{F} := \mathcal{F}_P(G)$. Then S is neither $\mathcal{F}$ -centralized nor $\mathcal{F}$ -normalized since S is $\mathcal{F}$ -conjugate to $Z(P) = \langle (13)(24) \rangle$.

Theorem 2.6. *The following assertions for a fusion system $\mathcal{F}$ on P are equivalent:*

- (1) (ROBERTS–SHPECTOROV) *Every subgroup of P is $\mathcal{F}$ -conjugate to an automized, receptive subgroup.*
- (2) *P is automized and every subgroup of P is $\mathcal{F}$ -conjugate to a normalized, receptive subgroup.*
- (3) (STANCU) *P is automized and every normalized subgroup of P is receptive.*
- (4) (BROTO–LEVI–OLIVER) *Every normalized subgroup of P is centralized and automized and every centralized subgroup is receptive.*

Under these circumstances we call $\mathcal{F}$ saturated.

For a saturated fusion system $\mathcal{F}$ on P and $S \leq P$ we have

- (i) S is $\mathcal{F}$ -centralized if and only if S is $\mathcal{F}$ -receptive.
- (ii) S is $\mathcal{F}$ -normalized if and only if S is $\mathcal{F}$ -centralized and $\mathcal{F}$ -automized.

Theorem 2.7. *If $P \in \text{Syl}_p(G)$, then $\mathcal{F}_P(G)$ is saturated.*

Proof. We prove Theorem 2.6(1) for $\mathcal{F} := \mathcal{F}_P(G)$. Let $Q \leq P$ and $N_P(Q) \leq R \in \text{Syl}_p(N_G(Q))$. By Sylow's theorem, there exists $g \in G$ such that

$$T := {}^gQ \leq {}^gR \leq P.$$

Since ${}^gR \in \text{Syl}_p({}^gN_G(Q)) = \text{Syl}_p(N_G(T))$, we have ${}^gR = N_P(T)$ and T is $\mathcal{F}$ -automized.

Now let $\varphi: S \rightarrow T$ be an arbitrary isomorphism in $\mathcal{F}$. Then there exists $a \in G$ with $\varphi(s) = {}^as$ for all $s \in S$. For $x \in N_\varphi$ there exists $y \in N_P(T)$ such that

$$axa^{-1}t = (\varphi c_x \varphi^{-1})(t) = {}^yt$$

for all $t \in T$. Hence, $y^{-1}axa^{-1} \in C_G(T)$ and $axa^{-1} \in N_P(T)C_G(T)$. By definition, $N_\varphi \leq N_P(S)$ is a p -group and ${}^aN_\varphi$ is a p -subgroup of $N_P(T)C_G(T)$. Since $N_P(T)$ is a Sylow p -subgroup of $N_G(T) \geq N_P(T)C_G(T)$, there exist $h \in N_P(T)$ and $z \in C_G(T)$ with ${}^{hza}N_\varphi \leq N_P(T)$. Then also ${}^{za}N_\varphi \leq N_P(T) \leq P$. For $s \in S$ we have ${}^{za}s = {}^z\varphi(s) = \varphi(s)$. Hence, the conjugation with za is an extension of φ to N_φ in $\mathcal{F}$. Consequently, T is $\mathcal{F}$ -receptive. $\square$

Example 2.8. Let $|P| > p$. A theorem of Gaschütz asserts that P has an outer automorphism of p -power order. Hence, P is not automized in $\mathcal{U}(P)$ and $\mathcal{U}(P)$ is not saturated.

Theorem 2.9 (ROBINSON, LEARY–STANCU). *For every saturated fusion system $\mathcal{F}$ on P there exists an infinite group G with $P \in \text{Syl}_p(G)$ such that $\mathcal{F} = \mathcal{F}_P(G)$.*

Definition 2.10. A saturated fusion system $\mathcal{F}$ is called *exotic* if there is no *finite* group G with $P \in \text{Syl}_p(G)$ and $\mathcal{F} = \mathcal{F}_P(G)$.

Example 2.11.

- (i) For $p = 2$ the only known simple exotic fusion systems are defined on the Sylow 2-subgroups of $\text{Spin}_7(q) \cong 2.\Omega_7(q)$ where q is an odd prime power. These are called the *Solomon fusion systems*. For $q = 3$ we have $|P| = 2^{10}$.
- (ii) For $p > 2$ many families of exotic fusion systems have been discovered recently. For instance, Ruiz–Viruel constructed an exotic fusion system $\mathcal{F}$ on the extraspecial group P of order 7^3 with exponent 7 such that all non-trivial elements of P are $\mathcal{F}$ -conjugate.

Most of the fusion and transfer theorems for finite groups stated in Section 1 have been translated to fusion systems. For instance, a saturated fusion system $\mathcal{F}$ is trivial if and only if $\text{Aut}_{\mathcal{F}}(Q)$ is a p -group for every $Q \leq P$. This will be generalized in the next section. To state some more theorems, we need the following constructions.

Definition 2.12. Let $\mathcal{F}$ be a saturated fusion system on P and $Q \leq P$.

- The fusion system $\text{C}_{\mathcal{F}}(Q)$ on $\text{C}_P(Q)$ consists of the morphisms $\varphi: S \rightarrow T$ such that there exists a morphism $\psi: QS \rightarrow QT$ in $\mathcal{F}$ with $\psi_S = \varphi$ and $\psi_Q = \text{id}_Q$.
- The fusion system $\text{N}_{\mathcal{F}}(Q)$ on $\text{N}_P(Q)$ consists of the morphisms $\varphi: S \rightarrow T$ such that there exists a morphism $\psi: QS \rightarrow QT$ in $\mathcal{F}$ with $\psi_S = \varphi$ and $\psi(Q) = Q$.
- The fusion system $\text{QC}_{\mathcal{F}}(Q)$ on $\text{QC}_P(Q)$ consists of the morphisms $\varphi: S \rightarrow T$ such that there exists a morphism $\psi: QS \rightarrow QT$ in $\mathcal{F}$ with $\psi_S = \varphi$ and $\psi_Q \in \text{Inn}(Q)$.

Recall that every subgroup $Q \leq P$ is $\mathcal{F}$ -conjugate to an $\mathcal{F}$ -normalized subgroup. In this case, Puig has shown that $\text{C}_{\mathcal{F}}(Q)$, $\text{N}_{\mathcal{F}}(Q)$ and $\text{QC}_{\mathcal{F}}(Q)$ are saturated.

Example 2.13. Let $Q \leq P \in \text{Syl}_p(G)$ and $\mathcal{F} = \mathcal{F}_P(G)$. If Q is $\mathcal{F}$ -normalized, then $\text{C}_{\mathcal{F}}(Q) = \mathcal{F}_{\text{C}_P(Q)}(\text{C}_G(Q))$, $\text{N}_{\mathcal{F}}(Q) = \mathcal{F}_{\text{N}_P(Q)}(\text{N}_G(Q))$ and $\text{QC}_{\mathcal{F}}(Q) = \mathcal{F}_{\text{QC}_P(Q)}(\text{QC}_G(Q))$.

Theorem 2.14 (KESSAR–LINCKELMANN). *A saturated fusion system $\mathcal{F}$ on P with $p > 2$ is trivial if and only if $\text{N}_{\mathcal{F}}(\text{Z}(\text{J}(P)))$ is trivial.*

Definition 2.15. For a saturated fusion system $\mathcal{F}$ on P we define

$$\begin{aligned} \text{Z}(\mathcal{F}) &:= \{x \in P : \varphi(x) = x \ \forall \varphi \in \text{Hom}_{\mathcal{F}}(\langle x \rangle, P)\} && (\text{center}), \\ \text{foc}(\mathcal{F}) &:= \langle \varphi(x)x^{-1} : x \in P, \ \varphi \in \text{Hom}_{\mathcal{F}}(\langle x \rangle, P) \rangle && (\text{focal subgroup}), \\ \text{hfp}(\mathcal{F}) &:= \langle \varphi(x)x^{-1} : x \in Q \leq P, \ \varphi \in \text{O}^p(\text{Aut}_{\mathcal{F}}(Q)) \rangle && (\text{hyperfocal subgroup}). \end{aligned}$$

Example 2.16.

- (i) The center $Z(\mathcal{F})$ is the largest subgroup $Q \leq P$ such that $C_{\mathcal{F}}(Q) = \mathcal{F}$.
- (ii) One can show that $\text{foc}(\mathcal{F}) = \text{hnp}(\mathcal{F})P'$ and $\text{foc}(\mathcal{F}) \cap Z(\mathcal{F}) = P' \cap Z(\mathcal{F})$. In particular, the *Fitting decomposition* $P = Z(\mathcal{F}) \times \text{foc}(\mathcal{F})$ holds whenever P is abelian.
- (iii) If $\mathcal{F} = \mathcal{F}_P(G)$, then $\text{foc}(\mathcal{F}) = \text{foc}_G(P)$, $\text{hnp}(\mathcal{F}) = \text{hnp}_G(P)$ and $Z(\mathcal{F}) \cong Z(G/O_{p'}(G))$ by the Z^* -theorem.

Theorem 2.17 (DÍAZ–GLESSER–PARK–STANCU). *Let $\mathcal{F}$ be a saturated fusion system on P .*

- (i) *If $\mathcal{E} \subseteq \mathcal{F}$ is a saturated subsystem (subcategory) on P , then $\text{foc}(\mathcal{F}) = \text{foc}(\mathcal{E}) \iff \text{hnp}(\mathcal{F}) = \text{hnp}(\mathcal{E})$.*
- (ii) *If P has no quotient isomorphic to $C_p \wr C_p$, then $\text{foc}(\mathcal{F}) = \text{foc}(\text{N}_{\mathcal{F}}(P))$. In particular, $\mathcal{F}$ is trivial if and only if $\text{Aut}_{\mathcal{F}}(P) = \text{Inn}(P)$.*

Theorem 2.18 (DÍAZ–GLESSER–MAZZA–PARK). *Let $\mathcal{F}$ be a saturated fusion system on P with $p \geq 5$. Then $\text{foc}(\mathcal{F}) = \text{foc}(\text{N}_{\mathcal{F}}(K))$ where K is the characteristic subgroup from Theorem 1.18.*

Kessar–Linckelmann and Onofrei–Stancu have translated Theorems 1.9 and 1.11 to fusion systems, but this requires the definition of $Qd(p)$ -free fusion systems.

3 Classification of fusion systems

Let $\mathcal{F}$ be a saturated fusion system on a finite p -group P . Let $\text{Out}_{\mathcal{F}}(Q) := \text{Aut}_{\mathcal{F}}(Q)/\text{Inn}(Q)$ for $Q \leq P$.

Theorem 3.1 (GLAUBERMAN–THOMPSON). *If $\text{foc}(\mathcal{F}) = P \neq 1$ and $p \geq 5$, then $\text{Out}_{\mathcal{F}}(P) \neq 1$.*

Definition 3.2. A subgroup $Q \leq P$ is called $\mathcal{F}$ -essential if

- $C_P(Q) \leq Q$,
- Q is $\mathcal{F}$ -normalized,
- there exists a *strongly p -embedded* subgroup $H < \text{Out}_{\mathcal{F}}(Q)$, i.e. $p \mid |H|$ and $p \nmid |H \cap {}^x H|$ for every $x \in \text{Out}_{\mathcal{F}}(Q) \setminus H$ (cf. Frobenius complement).⁸

Example 3.3.

- (i) Every $\mathcal{F}$ -essential subgroup $Q \leq P$ is $\mathcal{F}$ -radical, i.e. $O_p(\text{Aut}_{\mathcal{F}}(Q)) = \text{Inn}(Q)$. To prove this, let $H < U := \text{Out}_{\mathcal{F}}(Q)$ be strongly p -embedded. Let $H_p \leq U_p$ be Sylow p -subgroups of H and U respectively. For $x \in N_{U_p}(H_p)$, we have $1 \neq H_p \leq H \cap {}^x H$ and therefore $x \in N_{H_p}(H_p) = H_p$. Hence, $N_{U_p}(H_p) = H_p$ and $H_p = U_p$ by standard group theory. It follows that $O_p(U) = \bigcap_{x \in U} {}^x U_p \leq H \cap {}^u H$ for any $u \in U \setminus H$. So $O_p(U) = 1$.
- (ii) Part (i) shows that every essential subgroup Q has non-trivial p' -automorphisms and $\text{Out}_{\mathcal{F}}(Q)$ acts faithfully on $Q/\Phi(Q) \cong C_p^r$. Therefore, $\text{Out}_{\mathcal{F}}(Q) \leq \text{GL}(r, p)$.

⁸A finite group L contains a strongly p -embedded subgroup if and only if the graph with vertex set $\text{Syl}_p(L)$ and edges $(S, T) \iff S \cap T \neq 1$ is disconnected.

- (iii) Since P is $\mathcal{F}$ -automized, $\text{Out}_{\mathcal{F}}(P)$ is a p' -group and P is not essential.
- (iv) If P is abelian, then there are no essential subgroups, since P is the only self-centralizing subgroup.
- (v) Let $G = S_4$, $P \in \text{Syl}_2(G)$ and $\mathcal{F} = \mathcal{F}_P(G)$. Then $V_4 := \langle (12)(34), (13)(24) \rangle \leq P$ is $\mathcal{F}$ -essential since $\text{Out}_{\mathcal{F}}(V_4) = G/V_4 \cong S_3$ contains the strongly 2-embedded subgroup $P/V_4 \cong C_2$. On the other hand, $Q := \langle (12), (34) \rangle \cong V_4$ is not $\mathcal{F}$ -essential (provided $Q \leq P$).

Theorem 3.4 (ALPERIN–GOLDSCHMIDT’S fusion theorem). *Let $\mathcal{E}$ be a set of representatives for the $\mathcal{F}$ -conjugacy classes of essential subgroups. Every isomorphism in $\mathcal{F}$ is a composition of isomorphisms of the form $\varphi: S \rightarrow T$ with the following properties:*

- (i) $S, T \leq Q \in \mathcal{E} \cup \{P\}$.
- (ii) $\exists \psi \in \text{Aut}_{\mathcal{F}}(Q)$ such that $\psi_S = \varphi$,
- (iii) If $Q \in \mathcal{E}$, then ψ is a p -element.

The number $|\mathcal{E}|$ in Theorem 3.4 is called the *essential rank* of $\mathcal{F}$.

Theorem 3.5. *A group G contains a strongly p -embedded subgroup if and only if one of the following holds:*

- (1) $\text{O}_p(G) = 1$ and the Sylow p -subgroups of G are (non-trivial) cyclic or (generalized) quaternion 2-groups.
- (2) $\text{O}_{p'}(G/\text{O}_{p'}(G))$ is one of the following:
 - $\text{PSL}(2, p^n)$ for $n \geq 2$,
 - $\text{PSU}(3, p^n)$ for $n \geq 1$ and $(p, n) \neq (2, 1)$,
 - $\text{Sz}(2^{2n+1})$ for $p = 2$ and $n \geq 1$,
 - ${}^2G_2(3^{2n-1})$ for $p = 3$ and $n \geq 1$,
 - A_{2p} for $p \geq 5$,
 - $\text{PSL}_3(4)$, M_{11} for $p = 3$,
 - $\text{Aut}(\text{Sz}(32))$, ${}^2F_4(2)'$, McL , Fi_{22} for $p = 5$,
 - J_4 for $p = 11$.

Proof. The proof of $p = 2$ is due to Bender, while the case $p > 2$ was established during the CFSG. $\square$

Example 3.6.

- (i) In the situation of Theorem 3.5(1), every $P \in \text{Syl}_p(G)$ has a unique subgroup $\Omega(P)$ of order p . It is easy to see that $N_G(\Omega(P))$ is strongly p -embedded in G .
- (ii) The groups in Theorem 3.5(2) apart from A_{2p} , ${}^2G_2(3) \cong \text{PSL}(2, 8).3$ and $\text{Aut}(\text{Sz}(32)) \cong \text{Sz}(32).5$ are precisely the simple groups G with a non-cyclic *trivial intersection* (TI) Sylow p -subgroup P , i.e. $P \cap {}^gP = 1$ for all $g \in G \setminus N_G(P)$. Thus, $N_G(P)$ is strongly p -embedded in this case.
- (iii) Let $p \geq 5$ and $G = A_{2p}$. Then $H := G \cap (S_p \wr C_2)$ is strongly p -embedded in G .

Corollary 3.7. *Let $Q \leq P$ be $\mathcal{F}$ -essential with $p \geq 5$. Then one of the following holds for $N := N_P(Q)/Q$:*

- (1) N is cyclic, elementary abelian or 5_-^{1+2} (extraspecial of exponent 25).
- (2) $\exp(N) = p$ and $Z(N) = N' = \Phi(N) \cong C_p^n$ where $|N| = p^{3n}$ (i. e. N is special).

Alperin–Goldschmidt’s fusion theorem and Theorem 3.5 make it feasible to determine all saturated fusion systems on a given p -group. Parker–Semeraro have developed a MAGMA algorithm for this purpose and discovered fusion systems overlooked in previous work.⁹ Since “most” p -groups do not have non-trivial p' -automorphisms, there are very few essential subgroups and “most” fusion systems are trivial.

Definition 3.8.

- We call $\mathcal{F}$ *controlled* if there are no essential subgroups.
- We call P *resistant*¹⁰ if every fusion system on P is controlled.
- We call P *fusion-trivial* if every fusion system on P is trivial.

Example 3.9.

- (i) Let $P \in \text{Syl}_p(G)$. Then $\mathcal{F}_P(G)$ is controlled if and only if $N_G(P)$ controls fusion in P .
- (ii) By the Schur–Zassenhaus theorem, $\text{Inn}(P)$ has a complement A in $\text{Aut}_{\mathcal{F}}(P)$ since P is automized. If $\mathcal{F}$ is controlled, then $\mathcal{F} = \mathcal{F}_P(P \rtimes A)$. In particular, $\mathcal{F}$ is not exotic.
- (iii) Every abelian p -group is resistant by Example 3.3.
- (iv) Stancu proved that every metacyclic p -group for $p > 2$ is resistant. I proved that metacyclic 2-groups apart from D_{2^n} , Q_{2^n} , SD_{2^n} and $C_{2^n}^2$ are fusion-trivial.
- (v) Every 2-group of the form $C_{2^{a_1}} \times \dots \times C_{2^{a_n}}$ with $a_1 < \dots < a_n$ is fusion-trivial. The smallest non-trivial fusion-trivial p -group of odd order is `SmallGroup`(3⁶, 46).
- (vi) Let $\mathcal{F}$ be a saturated fusion system on $P = \langle x, y : x^4 = y^2 = 1, {}^y x = x^{-1} \rangle \cong D_8$. There are three cases:
 - (a) $\mathcal{F}$ is controlled and therefore trivial since $\text{Aut}(P) \cong D_8$ is a 2-group.
 - (b) There is exactly one essential subgroup, say $\langle x^2, y \rangle$. Then $\mathcal{F} = \mathcal{F}_P(S_4)$.
 - (c) There are two essential subgroups $\langle x^2, y \rangle$ and $\langle x^2, xy \rangle$. Then $\mathcal{F} = \mathcal{F}_P(\text{GL}(3, 2))$. In contrast to S_4 , all involutions in $\text{GL}(3, 2)$ are conjugate, namely to the rational canonical form

$$\begin{pmatrix} 1 & . & . \\ . & . & 1 \\ . & 1 & . \end{pmatrix}.$$

Definition 3.10. We call $Q \trianglelefteq P$ *normal* in $\mathcal{F}$ (and write $Q \trianglelefteq \mathcal{F}$) if $N_{\mathcal{F}}(Q) = \mathcal{F}$.

⁹<https://github.com/chris1961parker/fusion-systems>

¹⁰sometimes called *Swan group*

Let $Q, R \trianglelefteq \mathcal{F}$ and $\varphi \in \text{Hom}_{\mathcal{F}}(S, T)$. Then there exist $\psi \in \text{Hom}_{\mathcal{F}}(RS, RT)$ and $\tau \in \text{Hom}_{\mathcal{F}}(QRS, QRT)$ such that $\psi(R) = R$, $\psi_S = \varphi$, $\tau(Q) = Q$ and $\tau_{RS} = \psi$. Hence, $\tau(QR) = \tau(Q)\psi(R) = QR$ and $\tau_S = \psi_S = \varphi$. This shows that $\varphi \in N_{\mathcal{F}}(QR)$ and $QR \trianglelefteq \mathcal{F}$. The following definition is therefore justified.

Definition 3.11.

- The (unique) largest normal subgroup of $\mathcal{F}$ is denoted by $O_p(\mathcal{F})$.
- We call $\mathcal{F}$ *constrained* if $C_P(O_p(\mathcal{F})) \leq O_p(\mathcal{F})$.

Example 3.12.

- (i) If $Q \leq P \in \text{Syl}_p(G)$ and $Q \trianglelefteq G$, then $Q \trianglelefteq \mathcal{F}_P(G)$. On the other hand, if P is abelian, then $P \trianglelefteq \mathcal{F}_P(G)$, but not necessarily $P \trianglelefteq G$.
- (ii) Every essential subgroup contains $O_p(\mathcal{F})$ and $Z(\mathcal{F}) \leq O_p(\mathcal{F})$.
- (iii) Every controlled fusion system $\mathcal{F}$ on P is constrained with $O_p(\mathcal{F}) = P$. On the other hand, $\mathcal{F} := \mathcal{F}_{D_8}(S_4)$ is constrained with $O_2(\mathcal{F}) = V_4$, but not controlled.
- (iv) Let $G = \text{GL}(3, 2)$ and $P \in \text{Syl}_2(G)$. Then $\mathcal{F}_P(G)$ is not constrained since the two essential subgroups intersect in $Z(P)$ (cf. Example 3.9(vi)). Moreover, $\text{foc}(\mathcal{F}) = P \not\trianglelefteq \mathcal{F}$.
- (v) A group G is called *p-constrained* if $C_{\overline{G}}(O_p(\overline{G})) \leq O_p(\overline{G})$ where $\overline{G} := G/O_{p'}(G)$. In this case $\mathcal{F} := \mathcal{F}_P(G)$ is constrained with $\overline{O_p(\mathcal{F})} = \overline{O_p(\overline{G})}$. By Theorem 3.13 below every constrained fusion system arises in this way. The Hall–Higman lemma asserts that every (p) -solvable group is *p-constrained*.

Theorem 3.13 (Model theorem). *For every constrained fusion system $\mathcal{F}$ on P there exists a unique finite group G (called model) such that*

- (i) $P \in \text{Syl}_p(G)$ and $\mathcal{F} = \mathcal{F}_P(G)$.
- (ii) $O_{p'}(G) = 1$ and $C_G(O_p(G)) \leq O_p(G)$.

In particular, $\mathcal{F}$ is not exotic.

Let G be a model for the constrained fusion system $\mathcal{F}$ on P with $|P| = p^n$. A theorem of Hall shows that

$$|G| \leq |G/O_p(G)||P| \leq |\text{Aut}(O_p(G))|p^n \leq |\text{GL}(n, p)|p^n = (p^n - 1) \dots (p^n - p^{n-1})p^n.$$

In particular, there are only finitely many choices when P is given.

Example 3.14. If $\mathcal{F}$ is controlled, then $P \rtimes A$ is the model for $\mathcal{F}$ where $A \cong \text{Out}_{\mathcal{F}}(P)$ as in Example 3.9.

Theorem 3.15 (GLESSER). *Let $p > 2$ and $\mathcal{F}$ a non-trivial fusion system on P . Then $\mathcal{F}$ contains (as a subcategory) a non-trivial constrained fusion system on P .*

One can use Theorem 3.15 and the model theorem to decide whether a given group P is fusion-trivial. The fusion system $\mathcal{F}_{D_{16}}(\text{PGL}(2, 7))$ (found by Craven) shows that Glesser’s theorem fails for $p = 2$. In order to classify non-constrained fusion systems (especially exotic fusion systems), Oliver has introduced *reduced* and *tame* fusion systems. In an ongoing effort to simplify the CFSG, Aschbacher has investigated *simple* fusion systems. Unfortunately, fusion systems of simple groups are not always simple, but well-studied nevertheless.

4 Representation theory

Let F be an algebraically closed field of characteristic $p > 0$. Let B be a (p) -block of FG , i.e. an indecomposable direct summand. We fix a defect group $D \leq G$ of B .

Definition 4.1 (ALPERIN–BROUÉ, OLSSON).

- We call (Q, b_Q) a B -subpair if $Q \leq D$ and b_Q is a Brauer correspondent of B in $QC_G(Q)$, i.e. $b_Q^G = B$. For subpairs we write $(S, b_S) \trianglelefteq (T, b_T)$ if $S \leq T$ and $b_S^{TC_G(S)} = b_T^{TC_G(S)}$.¹¹ Let $\leq$ be the transitive closure of $\trianglelefteq$, i.e.

$$(S, b_S) \leq (T, b_T) \iff (S, b_T) = (T_1, b_1) \trianglelefteq \dots \trianglelefteq (T_n, b_n) = (T, b_T).$$

- We fix a B -subpair (D, b_D) (by Brauer's extended first main theorem, (D, b_D) is unique up to conjugation). It can be shown that for every $Q \leq D$ there exists a unique subpair of the form $(Q, b_Q) \leq (D, b_D)$. We fix those in the following. The fusion system $\mathcal{F} = \mathcal{F}_D(B)$ on D is defined by

$$\text{Hom}_{\mathcal{F}}(S, T) := \{\varphi: S \rightarrow T : \exists g \in G : {}^g(S, b_S) \leq (T, b_T) \wedge \varphi(s) = {}^g s \forall s \in S\}.$$

Theorem 4.2 (PUIG). *The fusion system $\mathcal{F}_D(B)$ is saturated.*

We call B *nilpotent* (*controlled*, *constrained*) if $\mathcal{F}_D(B)$ is trivial (controlled, constrained). The irreducible ordinary and modular characters of G can be distributed into blocks. We set $k(B) := |\text{Irr}(B)|$ and $l(B) := |\text{IBr}(B)|$. Moreover, let $\text{foc}(B) := \text{foc}(\mathcal{F}_D(B))$.

Example 4.3.

- The *principal* block $B = B_0(G)$ contains the trivial character of G . In this case $D \in \text{Syl}_p(G)$ and $\mathcal{F}_D(B) = \mathcal{F}_D(G)$. In particular, G is p -nilpotent if and only if B is nilpotent. In this case, all blocks of G are nilpotent.
- If $C_G(\text{O}_p(G)) \leq \text{O}_p(G)$, then $B_0(G)$ is the only block of G .
- In the context Definition 4.1, $\text{Out}_{\mathcal{F}}(D) = N_G(D, b_D)/DC_G(D)$ is called the *inertial quotient* of B and its order is the *inertial index*, which is coprime to p by Theorem 4.2.
- The dihedral group $G = D_{24}$ has a nilpotent 3-block with defect group $D \cong C_3$, while the principal 3-block is not nilpotent. This shows that D alone does not determine the fusion system of a block.

Conjecture 4.4. *For every block B of G with defect group D there exists a finite group H such that $D \in \text{Syl}_p(H)$ and $\mathcal{F}_D(B) = \mathcal{F}_D(H)$.*

Theorem 4.5.

- Let B be a block of S_n with defect group D . Then there exists an integer $w \geq 0$ (called the weight of B) such that $D \in \text{Syl}_p(S_{pw})$ and $\mathcal{F}_D(B) = \mathcal{F}_D(S_{pw})$.
- Let B be a block of A_n with defect group D . Then $\mathcal{F}_D(B) \in \{\mathcal{F}_D(S_{pw}), \mathcal{F}_D(A_{pw})\}$ for some $w \geq 0$.

¹¹Alperin–Broué require additionally that b_S is T -invariant, but Olsson showed that this is unnecessary.

Theorem 4.6 (HUMPHREYS, AN–DIETRICH). *Let B be a block of a group G of Lie type in characteristic p with defect group D . Then $D = 1$ or $D \in \text{Syl}_p(G)$ and $\mathcal{F}_D(B) = \mathcal{F}_D(G)$.*

It has been shown that there is no block with the exotic fusion systems mentioned in Example 2.11.

Theorem 4.7 (PUIG). *Let B be nilpotent. Then $B \cong (FD)^{n \times n}$ for some $n \geq 1$. In particular, B and FD are Morita equivalent, i. e. they have equivalent module categories. Moreover, $k(B) = k(D)$ and $l(B) = 1$.*

Theorem 4.8 (FONG–REYNOLDS). *Let b be a block of $N \trianglelefteq G$ with inertial group G_b . Then the Brauer correspondence $C \mapsto C^G$ gives a bijection between the blocks of G_b covering b and the blocks of G covering b . Moreover, C and C^G are Morita equivalent and have the same fusion system.*

Theorem 4.9 (Second Fong Reduction). *Let B be a block of G covering a G -invariant block of $N \trianglelefteq G$ with defect 0. Then B is Morita equivalent to a block of a finite group H with the same fusion system. Moreover, there exists a cyclic p' -subgroup $Z \leq Z(H)$ such that $H/Z \cong G/N$.*

The block of H in the situation of Theorem 4.9 is Morita equivalent to a twisted group algebra $F_\alpha[G/N]$ where $\alpha \in H^2(G/N, F^\times)$. Conversely, every such twisted group algebra is Morita equivalent to a block of a suitable central extension. If B is the principal block or if G/N has trivial Schur multiplier, then $\alpha = 1$ and B is Morita equivalent to $F[G/N]$. This applies also to the following two theorems.

Theorem 4.10 (KÜLSHAMMER). *If $D \trianglelefteq G$, then B is controlled and Morita equivalent to a twisted group algebra $F_\alpha[D \rtimes \text{Out}_{\mathcal{F}}(D)]$ where $\alpha \in H^2(\text{Out}_{\mathcal{F}}(D), F^\times)$.*

Theorem 4.11 (KÜLSHAMMER). *If G is p -solvable, then B is constrained and Morita equivalent to $F_\alpha H$ where H is the model for $\mathcal{F}_D(B)$ from Theorem 3.13 and $\alpha \in H^2(H, F^\times)$.*

Theorem 4.12 (EATON–KESSAR–KÜLSHAMMER–SAMBALE). *Every 2-block B with a metacyclic defect group D belongs to one of the following cases:*

- (1) B is nilpotent.
- (2) D is dihedral, semidihedral or quaternion and B has tame representation type (Morita equivalence classes classified up to scalars).
- (3) $D \cong C_{2^n}^2$ and B is Morita equivalent to $F[D \rtimes C_3]$.
- (4) $D \cong C_2^2$ and B is Morita equivalent to $B_0(A_5)$.

Conjecture 4.13 (Blockwise Z^* -conjecture). *Let B be a block with fusion system $\mathcal{F}$ and $Z := Z(\mathcal{F})$. Then B is Morita equivalent to its Brauer correspondent b_Z in $C_G(Z)$.*

Since $N_G(D, b_D) \leq C_G(Z)$, b_Z is indeed the unique Brauer correspondent of B by Brauer's first main theorem. Conjecture 4.13 holds for principal blocks by Example 2.16.

Theorem 4.14 (KÜLSHAMMER–OKUYAMA, WATANABE). *In the situation of Conjecture 4.13 we have $k(B) \geq k(b_Z)$ and $l(B) \geq l(b_Z)$ with equality in both cases if D is abelian.*

Conjecture 4.15 (ROUQUIER). *If $Q := \text{hyp}(\mathcal{F}_D(B))$ is abelian, then B is derived equivalent to its Brauer correspondent B_Q in $N_G(Q)$.*

Example 4.16. Suppose that B has abelian defect group D . Broué's conjecture predicts that B and B_Q are derived equivalent to their common Brauer correspondent in $N_G(D)$. This implies Rouquier's conjecture for B . Conversely, if Rouquier's conjecture and the blockwise Z^* -conjecture hold for B , then B is derived equivalent to its Brauer correspondent in $N_G(Q, b_Q) \cap C_G(Z) = N_G(D, b_D)$ since $D = Q \times Z$ by the Fitting decomposition (Example 2.16). Thus, Broué's conjecture holds for B .

Theorem 4.17 (WATANABE). *If Q is cyclic in the situation of Rouquier's conjecture, then $\mathcal{F}$ is controlled with $\text{Out}_{\mathcal{F}}(D) \leq C_{p-1}$ and*

$$\begin{aligned} k(B) &= k(B_Q) = k(D \rtimes \text{Out}_{\mathcal{F}}(D)), \\ l(B) &= l(B_Q) = |\text{Out}_{\mathcal{F}}(D)|. \end{aligned}$$

If $p > 2$ and D is non-abelian metacyclic, then Theorem 4.17 applies.

Definition 4.18. Let $\mathcal{F}$ be a saturated fusion system on P and $Q \trianglelefteq \mathcal{F}$. Then the (saturated) fusion system $\mathcal{F}/Q$ on P/Q consists of the morphism $\varphi: S/Q \rightarrow T/Q$ such that there exists a morphism $\psi: S \rightarrow T$ in $\mathcal{F}$ with $\varphi(xQ) = \psi(x)Q$ for all $x \in S$.

Theorem 4.19. *Let B be a block of G with defect group D and $\mathcal{F} = \mathcal{F}_D(B)$. Let (Q, b_Q) be a B -subpair such that Q is $\mathcal{F}$ -normalized. Then*

- (i) b_Q has defect group $QC_D(Q)$ and fusion system $QC_{\mathcal{F}}(Q)$.
- (ii) $b_Q^{\text{N}_G(Q)}$ has defect group $N_D(Q)$ and fusion system $N_{\mathcal{F}}(Q)$.
- (iii) b_Q dominates a unique block $\overline{b_Q}$ of $C_G(Q)Q/Q$ with defect group $C_D(Q)Q/Q$ and fusion system $QC_{\mathcal{F}}(Q)/Q$. Moreover, $l(b_Q) = l(\overline{b_Q})$.

In the situation of Theorem 4.19 the map $S \mapsto S/Q$ is a bijection between the set of $QC_{\mathcal{F}}(Q)$ -essential subgroups and the set of $QC_{\mathcal{F}}(Q)/Q$ -essential subgroups. This allows inductive arguments.

Theorem 4.20 (BRAUER). *Let B be a block of G with defect group D and $\mathcal{F} = \mathcal{F}_D(B)$. Let $\mathcal{X} \subseteq D$ be a set of representatives for the $\mathcal{F}$ -conjugacy classes of D such that $\langle x \rangle$ is $\mathcal{F}$ -normalized for $x \in \mathcal{X}$. Then*

$$k(B) = \sum_{x \in \mathcal{X}} l(b_x) = \sum_{x \in \mathcal{X}} l(\overline{b_x}),$$

where $b_x := b_{\langle x \rangle}$. In particular, $k(B) - l(B)$ is locally determined.

The fusion system of a block does not determine $k(B)$ or $l(B)$. For example, the group

$$G = \text{SmallGroup}(72, 23) \cong C_3^2 \rtimes D_8$$

with $|Z(G)| = 2$ from the *small groups library* has two 3-blocks B_0, B_1 with defect group $D = C_3^2$ and fusion system $\mathcal{F}_D(S_3^2)$, but $l(B_0) = 4$ and $l(B_1) = 1$. We need an additional ingredient: For an F -algebra A let $z(A)$ be the number of simple projective A -modules up to isomorphism.

Conjecture 4.21 (ALPERIN’s weight conjecture). *Let B be a block of G with defect group D and $\mathcal{F} = \mathcal{F}_D(B)$. Let $\mathcal{R}$ be a set of representatives for the $\mathcal{F}$ -conjugacy classes of self-centralizing, $\mathcal{F}$ -centralized subgroups of D . Then*

$$l(B) = \sum_{Q \in \mathcal{R}} z(F_{\gamma_Q} \text{Out}_{\mathcal{F}}(Q))$$

where $\gamma_Q \in H^2(\text{Out}_{\mathcal{F}}(Q), F^\times)$ is the so-called Külshammer–Puig class.

Example 4.22.

- (i) Suppose that B is controlled in the situation of Conjecture 4.21. Then $z(F_{\gamma_Q} \text{Out}_{\mathcal{F}}(Q)) = 0$ for $Q < D$, since $N_D(Q)/Q$ is a non-trivial normal p -subgroup of $\text{Out}_{\mathcal{F}}(Q)$. Hence, Alperin’s conjecture becomes $l(B) = z(F_{\gamma_D} \text{Out}_{\mathcal{F}}(D))$. If in addition B is the principal block (or $\text{Out}_{\mathcal{F}}(D)$ has trivial Schur multiplier), then $l(B) = z(F \text{Out}_{\mathcal{F}}(D)) = k(\text{Out}_{\mathcal{F}}(D))$.
- (ii) Let B be the principal 2-block of S_4 with $D = \langle x, y \rangle$ as in Example 3.9. The self-centralizing, $\mathcal{F}$ -centralized subgroups are $Q_1 = \langle x^2, y \rangle$, $Q_2 = \langle x^2, xy \rangle$, $Q_3 = \langle x \rangle$ and $Q_4 = D$. Alperin’s conjecture becomes

$$l(B) = \sum_{i=1}^4 z(F_{\gamma_{Q_i}} \text{Out}_{\mathcal{F}}(Q_i)) = z(FS_3) + 2z(FC_2) + z(F) = 1 + 0 + 1 = 2.$$

Definition 4.23. The *height* $h \geq 0$ of $\chi \in \text{Irr}(B)$ is defined by $\chi(1)_p = p^h |G : D|_p$. Let $k_h(B)$ be the number of $\chi \in \text{Irr}(B)$ with height h .

Theorem 4.24 (BROUÉ–PUIG, ROBINSON). *Let B be a block with defect group D . Then*

- (i) $|D/\text{foc}(B)|$ divides $k_0(B)$ with equality if and only if B is nilpotent.
- (ii) $|Z(D)\text{foc}(B)/\text{foc}(B)|$ divides $k_h(B)$ for all $h \geq 0$.

If D is abelian, then $|Z(\mathcal{F}_D(B))|$ divides $k(B)$, because $D = \text{foc}(B) \times Z(\mathcal{F}_D(B))$.

Dade’s conjecture, expressing $k_h(B)$ in terms of alternating sums, has been reformulated in terms of fusion systems by Robinson (*ordinary weight conjecture*). Kessar–Linckelmann–Lynd–Semeraro have generalized this and other conjectures in block theory to statements on abstract fusion systems.

References

- [1] J. L. Alperin, *Sylow intersections and fusion*, J. Algebra **6** (1967), 222–241.
- [2] J. L. Alperin and M. Broué, *Local methods in block theory*, Ann. of Math. (2) **110** (1979), 143–157.
- [3] J. An and H. Dietrich, *The AWC-goodness and essential rank of sporadic simple groups*, J. Algebra **356** (2012), 325–354.
- [4] J. An and H. Dietrich, *The essential rank of fusion systems of blocks of symmetric groups*, Internat. J. Algebra Comput. **22** (2012), 1250002, 15.
- [5] J. An and H. Dietrich, *The essential rank of Brauer categories for finite groups of Lie type*, Bull. London Math. Soc. **45** (2013), 363–369.

- [6] K. K. S. Andersen, B. Oliver and J. Ventura, *Reduced, tame and exotic fusion systems*, Proc. Lond. Math. Soc. (3) **105** (2012), 87–152.
- [7] M. Aschbacher, *Classifying finite simple groups and 2-fusion systems*, ICCM Not. **3** (2015), 35–42.
- [8] M. Aschbacher, R. Kessar and B. Oliver, *Fusion systems in algebra and topology*, London Mathematical Society Lecture Note Series, Vol. 391, Cambridge University Press, Cambridge, 2011.
- [9] M. Aschbacher and B. Oliver, *Fusion systems*, Bull. Amer. Math. Soc. (N.S.) **53** (2016), 555–615.
- [10] S. Bayard and J. Lynd, *Realizing finite groups as automizers*, J. Group Theory **27** (2024), 713–726.
- [11] H. Bender, *Transitive Gruppen gerader Ordnung, in denen jede Involution genau einen Punkt festläßt*, J. Algebra **17** (1971), 527–554.
- [12] C. Broto, J. M. Möller and B. Oliver, *Equivalences between fusion systems of finite groups of Lie type*, J. Amer. Math. Soc. **25** (2012), 1–20.
- [13] M. Broué and L. Puig, *A Frobenius theorem for blocks*, Invent. Math. **56** (1980), 117–128.
- [14] D. A. Craven, *The theory of fusion systems*, Cambridge Studies in Advanced Mathematics, Vol. 131, Cambridge University Press, Cambridge, 2011.
- [15] A. Díaz, A. Glessner, N. Mazza and S. Park, *Control of transfer and weak closure in fusion systems*, J. Algebra **323** (2010), 382–392.
- [16] A. Díaz, A. Glessner, S. Park and R. Stancu, *Tate’s and Yoshida’s theorems on control of transfer for fusion systems*, J. Lond. Math. Soc. (2) **84** (2011), 475–494.
- [17] C. W. Eaton, R. Kessar, B. Külshammer and B. Sambale, *2-blocks with abelian defect groups*, Adv. Math. **254** (2014), 706–735.
- [18] R. J. Flores and R. M. Foote, *Strongly closed subgroups of finite groups*, Adv. Math. **222** (2009), 453–484.
- [19] G. Glauberman, *Central elements in core-free groups*, J. Algebra **4** (1966), 403–420.
- [20] G. Glauberman, *Subgroups of finite groups*, Bull. Amer. Math. Soc. **73** (1967), 1–12.
- [21] G. Glauberman, *A characteristic subgroup of a p -stable group*, Canad. J. Math. **20** (1968), 1101–1135.
- [22] A. Glessner, *Sparse fusion systems*, Proc. Edinb. Math. Soc. (2) **56** (2013), 135–150.
- [23] D. M. Goldschmidt, *A conjugation family for finite groups*, J. Algebra **16** (1970), 138–142.
- [24] D. M. Goldschmidt, *2-fusion in finite groups*, Ann. of Math. (2) **99** (1974), 70–117.
- [25] D. Gorenstein and R. Lyons, *The local structure of finite groups of characteristic 2 type*, Mem. Amer. Math. Soc. **42** (1983), 1–731.
- [26] G. T. Helleloid and U. Martin, *The automorphism group of a finite p -group is almost always a p -group*, J. Algebra **312** (2007), 294–329.
- [27] E. Henke and J. Semeraro, *Centralizers of normal subgroups and the Z^* -theorem*, J. Algebra **439** (2015), 511–514.
- [28] A. Jaber, *Block fusion systems over maximal nilpotency class 3-groups*, arXiv:2207.06454v1.
- [29] M. W. Jacobsen, *Block fusion systems of the alternating groups*, arXiv:1204.2702v1.
- [30] R. Kessar and M. Linckelmann, *ZJ-theorems for fusion systems*, Trans. Amer. Math. Soc. **360** (2008), 3093–3106.
- [31] R. Kessar, M. Linckelmann, J. Lynd and J. Semeraro, *Weight conjectures for fusion systems*, Adv. Math. **357** (2019), 106825, 40pp.
- [32] R. Kessar, M. Linckelmann and G. Navarro, *A characterisation of nilpotent blocks*, Proc. Amer. Math. Soc. **143** (2015), 5129–5138.

- [33] R. Kessar and R. Stancu, *A reduction theorem for fusion systems of blocks*, J. Algebra **319** (2008), 806–823.
- [34] B. Külshammer, *On p -blocks of p -solvable groups*, Comm. Algebra **9** (1981), 1763–1785.
- [35] B. Külshammer, *Crossed products and blocks with normal defect groups*, Comm. Algebra **13** (1985), 147–168.
- [36] B. Külshammer and T. Okuyama, *On centrally controlled blocks of finite groups*, unpublished.
- [37] I. J. Leary and R. Stancu, *Realising fusion systems*, Algebra Number Theory **1** (2007), 17–34.
- [38] M. Linckelmann, *Introduction to fusion systems*, in: Group representation theory, 79–113, EPFL Press, Lausanne, 2007. Revised version: <http://web.mat.bham.ac.uk/C.W.Parker/Fusion/fusion-intro.pdf>.
- [39] A. Nectoux, *Fusion systems of finite alternating groups*, PhD thesis, University of Auckland, 2015.
- [40] B. Oliver and A. Ruiz, *Simplicity of fusion systems of finite simple groups*, Trans. Amer. Math. Soc. **374** (2021), 7743–7777.
- [41] J. B. Olsson, *On subpairs and modular representation theory*, J. Algebra **76** (1982), 261–279.
- [42] S. Onofrei and R. Stancu, *A characteristic subgroup for fusion systems*, J. Algebra **322** (2009), 1705–1718.
- [43] S. Park, *Realizing fusion systems inside finite groups*, Proc. Amer. Math. Soc. **144** (2016), 3291–3294.
- [44] C. Parker and J. Semeraro, *Algorithms for fusion systems with applications to p -groups of small order*, Math. Comp. **90** (2021), 2415–2461.
- [45] L. Puig, *Sur un théorème d’Alperin*, C. R. Acad. Sci. Paris Sér. A **278** (1974), 1013–1016.
- [46] L. Puig, *Structure locale dans les groupes finis*, Bull. Soc. Math. France Suppl. Mém. (1976), 5–132.
- [47] L. Puig, *Nilpotent blocks and their source algebras*, Invent. Math. **93** (1988), 77–116.
- [48] L. Puig, *Frobenius categories*, J. Algebra **303** (2006), 309–357.
- [49] L. Puig, *Frobenius categories versus Brauer blocks*, Progress in Mathematics, Vol. 274, Birkhäuser Verlag, Basel, 2009.
- [50] L. Puig, *The Frobenius P -categories via the Alperin condition*, J. Algebra **324** (2010), 2923–2930.
- [51] K. Roberts and S. Shpectorov, *On the definition of saturated fusion systems*, J. Group Theory **12** (2009), 679–687.
- [52] G. R. Robinson, *Weight conjectures for ordinary characters*, J. Algebra **276** (2004), 761–775.
- [53] G. R. Robinson, *Amalgams, blocks, weights, fusion systems and finite simple groups*, J. Algebra **314** (2007), 912–923.
- [54] G. R. Robinson, *On the focal defect group of a block, characters of height zero, and lower defect group multiplicities*, J. Algebra **320** (2008), 2624–2628.
- [55] R. Rouquier, *Block theory via stable and Rickard equivalences*, in: Modular representation theory of finite groups (Charlottesville, VA, 1998), 101–146, de Gruyter, Berlin, 2001.
- [56] A. Ruiz and A. Viruel, *The classification of p -local finite groups over the extraspecial group of order p^3 and exponent p* , Math. Z. **248** (2004), 45–65.
- [57] B. Sambale, *Fusion systems on metacyclic 2-groups*, Osaka J. Math. **49** (2012), 325–329.
- [58] P. Serwene, *Reduction theorems for generalised block fusion systems*, J. Algebra **614** (2023), 458–480.
- [59] R. Solomon, *Finite groups with Sylow 2-subgroups of type .3*, J. Algebra **28** (1974), 182–198.
- [60] R. Stancu, *Control of fusion in fusion systems*, J. Algebra Appl. **5** (2006), 817–837.
- [61] B. Stellmacher, *A characteristic subgroup of Σ_4 -free groups*, Israel J. Math. **94** (1996), 367–379.

- [62] J. Thévenaz, *Most finite groups are p -nilpotent*, Exposition. Math. **11** (1993), 359–363.
- [63] R. W. van der Waall, *On p -nilpotent forcing groups*, Indag. Math. (N.S.) **2** (1991), 367–384.
- [64] A. Watanabe, *Note on a p -block of a finite group with abelian defect group*, Osaka J. Math. **26** (1989), 829–836.
- [65] A. Watanabe, *The number of irreducible Brauer characters in a p -block of a finite group with cyclic hyperfocal subgroup*, J. Algebra **416** (2014), 167–183.
- [66] W. Xiao, *Glauberman’s conjecture, Mazurov’s problem and Peng’s problem*, Sci. China, Ser. A **34** (1991), 1025–1031.
- [67] T. Yoshida, *Character-theoretic transfer*, J. Algebra **52** (1978), 1–38.